

# Volatility as an Asset Class: Variance Swaps, VIX Options, and the Volatility Risk Premium

Group 4

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## Abstract

This paper studies volatility as a tradable asset class through variance swaps, VIX derivatives, and the volatility risk premium. The project combines theoretical derivations, Monte Carlo simulation, Fourier-cosine pricing, and later empirical validation using market data. The current version focuses on the model structure, the Heston Monte Carlo framework, and the CIR-based COS pricing approach for VIX derivatives.

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# 1 Introduction

Volatility plays a central role in modern financial markets. In its most basic interpretation, volatility measures the uncertainty of asset returns. In derivative pricing, however, volatility is more than a statistical input: it becomes a risk factor, a market expectation, and in many cases a directly tradable quantity. Products such as variance swaps, VIX futures, and VIX options allow investors to take positions on future realised or implied volatility without necessarily taking a direct directional position in the underlying equity index.

The development of volatility-linked derivatives has changed the way market participants understand and manage risk. A variance swap, for example, provides exposure to realised variance over a fixed time interval. VIX futures and VIX options provide exposure to market expectations of future volatility. These instruments are widely used for hedging, speculation, relative value trading, and volatility risk premium strategies. Their pricing therefore requires a framework that connects stochastic variance dynamics, risk-neutral valuation, numerical approximation, and empirical validation.

The main objective of this project is to study volatility as an asset class through variance swaps and VIX-linked derivatives. The project develops a quantitative framework that combines theoretical derivations with numerical pricing methods. The theoretical part is based on a Heston-type stochastic volatility model, where the instantaneous variance follows a Cox–Ingersoll–Ross square-root process. This model is useful as a benchmark because it captures mean reversion in variance while preserving a level of analytical tractability.

The project is organised around four completed components. The first component reviews the literature on volatility derivatives, variance risk premia, stochastic volatility models, and recent numerical methods. The second component develops the mathematical framework, including the variance swap strike, the model-implied VIX proxy, and the characteristic function of terminal CIR variance. The third component implements a Monte Carlo engine for Heston simulation, realised variance estimation, variance swap validation, and model-implied VIX call pricing. The fourth component implements a Fourier-cosine pricing engine for VIX derivatives using the characteristic function of terminal variance.

The numerical part of the project is deliberately built around two complementary methods. Monte Carlo simulation is flexible and intuitive. It allows stock and variance paths to be generated directly and can be used for path-dependent quantities such as realised variance. Its main limitation is that it contains sampling error and may require many paths for stable estimates. The Fourier-cosine method, in contrast, is deterministic once the model parameters and truncation interval are fixed. It can be computationally efficient when the characteristic function of the relevant state variable is available. The

use of both approaches provides a natural cross-validation framework.

A key modelling distinction in this project is the difference between the official Cboe VIX index and the model-implied VIX proxy used in the numerical implementation. The official VIX is computed from a strip of out-of-the-money SPX option prices according to the Cboe methodology. The proxy used in this project is instead derived from the conditional expectation of future variance under the Heston model. Therefore, the project does not claim to reproduce the exchange methodology exactly. Rather, it studies a Heston-implied VIX-type quantity that is mathematically convenient for pricing and validation.

The research contribution of the current implementation is not a new stochastic volatility model, but a structured numerical framework that connects the theory and implementation of volatility-linked derivatives. The Monte Carlo engine provides a simulation-based estimate of realised variance and VIX option payoffs. The COS engine provides a characteristic-function-based benchmark. The comparison between the two engines helps verify whether the implementation is internally consistent.

The rest of the paper is structured as follows. Section 2 reviews the relevant literature and recent methodological developments. Section 3 presents the mathematical framework. Section 4 develops the Monte Carlo methodology. Section 5 presents the CIR-based COS pricing engine. The later sections discuss numerical validation, code architecture, limitations, and future extensions toward real data calibration and volatility risk premium backtesting.

## 2 Literature Review

The literature on volatility as an asset class can be organised around several connected themes: variance swap replication, stochastic volatility modelling, VIX derivatives, numerical pricing methods, and the empirical volatility risk premium. This section reviews these themes in relation to the current project.

### 2.1 Volatility Derivatives and Variance Swaps

Variance swaps are among the most direct instruments for trading volatility. Unlike a standard option, whose payoff depends on the terminal level of the underlying asset, a variance swap depends on realised variance over a specified time interval. The floating leg is determined by the realised variance of the underlying, while the fixed leg is set at inception. The fair variance strike is therefore the risk-neutral expectation of future realised variance.

This relationship makes variance swaps useful for both theoretical and practical reasons. Theoretically, they provide a clean connection between quadratic variation and

derivative pricing. Practically, they allow investors to trade future realised volatility without choosing a particular option strike. For this reason, variance swaps are often used as a benchmark instrument in volatility modelling.

In the Heston framework, the fair variance swap strike can be obtained from the expected integrated variance. Since the variance process is mean-reverting, the fair strike depends on the initial variance, long-run variance, mean-reversion speed, and maturity. This gives a closed-form expression that is useful for validating Monte Carlo simulation.

## 2.2 Stochastic Volatility and the Heston Benchmark

The Black–Scholes model assumes constant volatility, which is not sufficient to describe observed option markets. Market implied volatility varies across strikes and maturities, producing volatility smiles and surfaces. Stochastic volatility models address this limitation by allowing volatility or variance to evolve randomly over time.

The Heston model is one of the most important stochastic volatility benchmarks. It specifies a square-root process for variance and allows the asset price and variance shocks to be correlated. The variance process is mean-reverting, which is consistent with the empirical observation that periods of high volatility tend to be followed by gradual normalisation. The model also provides analytical and semi-analytical tools through characteristic functions, which makes it suitable for numerical pricing (Hirsa, 2024).

In the current project, the Heston model is used not because it is the most realistic model for all market conditions, but because it provides a tractable foundation for implementing and comparing numerical methods. It allows the project to derive a variance swap benchmark, simulate stochastic variance paths, construct a model-implied VIX proxy, and use the terminal CIR characteristic function in the COS method.

## 2.3 VIX Derivatives and Model-Implied VIX

The VIX index is widely interpreted as a market measure of expected future volatility. It is constructed from SPX option prices and is often referred to as a measure of market-implied volatility. VIX futures and VIX options are therefore important instruments for trading volatility expectations.

However, the official VIX methodology and a model-implied VIX proxy are not the same object. The official Cboe methodology constructs the index from a strip of out-of-the-money options and uses a specific interpolation and aggregation procedure (Cboe Global Markets, 2024). A stochastic volatility model, by contrast, may define a VIX-type quantity as the conditional expectation of future variance over a fixed time window. This produces a mathematically convenient model-implied proxy.

This distinction is important for the present project. The numerical implementation

prices derivatives on

$$\widetilde{\text{VIX}}_T = 100\sqrt{A + Bv_T},$$

where  $v_T$  is terminal Heston variance and  $A, B$  are obtained from the conditional expectation of future CIR variance. This proxy captures the model’s internal expectation of future variance, but it does not fully reproduce the market construction of the official VIX index.

## 2.4 Monte Carlo Methods in Computational Finance

Monte Carlo simulation is a standard method for derivative pricing when analytical solutions are unavailable or inconvenient. It approximates risk-neutral expectations by averaging simulated discounted payoffs. Its main advantage is flexibility: once a stochastic model is specified, one can simulate paths and compute many types of payoffs.

In this project, Monte Carlo simulation is used for three purposes. First, it simulates Heston stock and variance paths. Second, it estimates realised variance from simulated log returns. Third, it prices a model-implied VIX call by transforming terminal variance into a VIX level and averaging the discounted payoff.

The main limitation of Monte Carlo simulation is statistical error. The standard error decreases at the rate  $O(N^{-1/2})$ , where  $N$  is the number of simulated paths. Therefore, high accuracy may require a large number of paths. In addition, discretisation error arises because continuous-time stochastic differential equations are approximated on a finite time grid.

The simulation of square-root variance processes requires particular care near the zero boundary. Although the CIR process is non-negative in continuous time, discretisation schemes may produce small negative values. Modern discussions of Heston and square-root process simulation emphasise that numerical schemes can materially affect pricing accuracy, especially for variance-sensitive derivatives (Choi and Kwok, 2024; Abi Jaber, 2024). For this reason, the project uses a full-truncation Euler scheme to preserve numerical stability.

## 2.5 Fourier and COS Pricing Methods

Transform-based pricing methods are widely used in quantitative finance because characteristic functions are often available even when probability densities are difficult to use directly. The Fourier-cosine method belongs to this family. It approximates the density or pricing integral on a finite interval by a cosine expansion and then computes option prices from characteristic function values.

The COS method is particularly attractive for models with known characteristic functions, including affine stochastic volatility models. It can converge rapidly when the payoff

is sufficiently regular and the truncation interval is chosen appropriately. However, the method is sensitive to numerical choices. If the truncation interval is too narrow, relevant probability mass may be excluded. If it is too wide, convergence may become less efficient. Recent work on the COS method treats both the truncation range and the number of expansion terms as central accuracy controls rather than minor implementation details (Junike and Pankrashkin, 2022; Junike, 2024).

In the current project, the COS method is applied to terminal CIR variance. Since the model-implied VIX payoff is a function of terminal variance, the terminal CIR characteristic function can be used directly. The payoff coefficients are computed numerically because the VIX payoff contains a square-root transformation of variance. This implementation provides a deterministic benchmark for the Monte Carlo VIX call price.

## 2.6 Volatility Risk Premium and Future Extensions

The volatility risk premium refers to the difference between implied variance and expected realised variance under the physical measure. A positive premium means that market-implied variance exceeds expected future realised variance. This phenomenon is often interpreted as compensation demanded by investors for bearing volatility risk.

The current project prepares the numerical foundation for studying the volatility risk premium, but does not yet perform a full empirical backtest. Such a backtest would require historical VIX, SPX option, VIX futures, and realised variance data. It would also require careful distinction between risk-neutral expectations and physical expectations. Therefore, volatility risk premium analysis is treated as a future extension after the data collection and calibration stages.

## 2.7 Summary of Literature Implications

The reviewed literature suggests three important implications for the project. First, volatility-linked derivatives require models that treat variance as a dynamic state variable rather than a fixed parameter. Second, different numerical methods should be compared because each method has different sources of error. Third, the distinction between model-implied volatility quantities and market-defined indices must be maintained throughout the analysis.

These implications directly motivate the W3 and W4 components. The Monte Carlo engine provides a flexible simulation-based framework, while the COS engine provides a deterministic characteristic-function-based benchmark. Their comparison strengthens the internal validation of the project and prepares the framework for later empirical extensions.



### 3 Mathematical Framework

This section presents the mathematical framework used in the project. The objective is to define the stochastic volatility model, derive the variance swap benchmark, construct the model-implied VIX proxy, and introduce the characteristic function of terminal CIR variance. These results provide the theoretical basis for the Monte Carlo engine in Section 4 and the COS pricing engine in Section 5.

#### 3.1 Risk-Neutral Pricing Principle

Derivative prices in this project are computed under the risk-neutral measure  $\mathbb{Q}$ . If a derivative pays a terminal payoff  $\Phi$  at maturity  $T$ , its time-zero price is given by

$$V_0 = e^{-rT} \mathbb{E}^{\mathbb{Q}} [\Phi], \quad (1)$$

where  $r$  is the continuously compounded risk-free rate. This representation is the common foundation for both simulation-based and transform-based pricing. Monte Carlo approximates the expectation by averaging simulated payoffs, while the COS method approximates the same expectation using the characteristic function of the state variable.

In the present setting, the payoff may depend on realised variance, terminal variance, or a model-implied VIX level. The choice of state variable determines which numerical method is more convenient. Path-dependent quantities, such as realised variance, are naturally handled by Monte Carlo simulation. Payoffs depending on terminal variance are suitable for characteristic-function-based methods because terminal CIR variance has a known transform.

#### 3.2 Heston Stochastic Volatility Model

The project uses the Heston stochastic volatility model as the baseline framework. Under the risk-neutral measure  $\mathbb{Q}$ , the asset price  $S_t$  and instantaneous variance  $v_t$  satisfy

$$\frac{dS_t}{S_t} = r dt + \sqrt{v_t} dW_t^S, \quad (2)$$

$$dv_t = \kappa(\theta - v_t) dt + \sigma_v \sqrt{v_t} dW_t^v, \quad (3)$$

$$d\langle W^S, W^v \rangle_t = \rho dt. \quad (4)$$

Here,  $r$  is the risk-free rate,  $\kappa > 0$  is the speed of mean reversion,  $\theta > 0$  is the long-run variance level,  $\sigma_v > 0$  is the volatility of variance, and  $\rho \in [-1, 1]$  is the correlation between the stock and variance shocks.

The variance process in (3) is a Cox–Ingersoll–Ross process. Its drift term  $\kappa(\theta - v_t)$  pulls variance toward the long-run level  $\theta$ . If  $v_t > \theta$ , the drift is negative and variance

tends to decrease. If  $v_t < \theta$ , the drift is positive and variance tends to increase. This mean-reverting structure is important for the model-implied VIX term structure.

The diffusion term  $\sigma_v \sqrt{v_t} dW_t^v$  makes the variance stochastic. The square-root specification ensures that the diffusion coefficient decreases as variance approaches zero. In continuous time, the process is non-negative under standard parameter conditions. In numerical simulation, however, discrete approximations may still produce small negative values, which motivates the use of full-truncation Euler simulation in the Monte Carlo engine.

### 3.3 Conditional Mean of the CIR Variance Process

A central quantity used throughout the project is the conditional expectation of future variance. Taking expectation in the CIR variance equation gives the ordinary differential equation

$$\frac{d}{dt} \mathbb{E}^{\mathbb{Q}}[v_t] = \kappa (\theta - \mathbb{E}^{\mathbb{Q}}[v_t]). \quad (5)$$

Solving this equation with initial value  $v_0$  gives

$$\mathbb{E}^{\mathbb{Q}}[v_t] = \theta + (v_0 - \theta)e^{-\kappa t}. \quad (6)$$

More generally, conditional on information at time  $t$ ,

$$\mathbb{E}_t^{\mathbb{Q}}[v_{t+u}] = \theta + (v_t - \theta)e^{-\kappa u}. \quad (7)$$

This formula is used twice in the project. First, it gives the analytical variance swap strike. Second, it gives the affine representation of the model-implied VIX proxy.

### 3.4 Variance Swap Strike

A variance swap is a contract whose payoff depends on realised variance over a fixed interval. In continuous time, the idealised floating leg is based on average integrated variance:

$$\frac{1}{T} \int_0^T v_s ds. \quad (8)$$

The fair variance swap strike is therefore defined as the risk-neutral expectation of this quantity:

$$K_{\text{var}}(0, T) = \frac{1}{T} \mathbb{E}^{\mathbb{Q}} \left[ \int_0^T v_s ds \right]. \quad (9)$$

Using Fubini's theorem and the conditional mean in (6), we obtain

$$K_{\text{var}}(0, T) = \frac{1}{T} \int_0^T \mathbb{E}^{\mathbb{Q}}[v_s] ds \quad (10)$$

$$= \frac{1}{T} \int_0^T [\theta + (v_0 - \theta)e^{-\kappa s}] ds. \quad (11)$$

Evaluating the integral yields

$$K_{\text{var}}(0, T) = \theta + (v_0 - \theta) \frac{1 - e^{-\kappa T}}{\kappa T}. \quad (12)$$

This expression is important because it provides a closed-form benchmark. In the Monte Carlo part, the simulated realised variance is compared with (12). If the simulation is implemented correctly and the number of paths is sufficiently large, the Monte Carlo mean should be close to the analytical strike.

### 3.5 Realised Variance and Its Discrete Approximation

In the numerical implementation, realised variance is computed from simulated stock prices observed on a finite time grid

$$0 = t_0 < t_1 < \dots < t_m = T.$$

The annualised realised variance estimator is

$$\widehat{\text{RV}}_{0,T}^{(m)} = \frac{1}{T} \sum_{i=1}^m \left( \log \frac{S_{t_i}}{S_{t_{i-1}}} \right)^2. \quad (13)$$

This is the discrete analogue of quadratic variation. Under a continuous diffusion model, as the time mesh becomes finer, the sum of squared log returns approximates the integrated variance:

$$\widehat{\text{RV}}_{0,T}^{(m)} \rightarrow \frac{1}{T} \int_0^T v_s ds. \quad (14)$$

This connection justifies comparing simulated realised variance with the analytical variance swap strike. The realised variance estimator is path-based, while  $K_{\text{var}}$  is expectation-based. Therefore, the comparison is performed after averaging realised variance across many Monte Carlo paths.

### 3.6 Model-Implied VIX Proxy

The official VIX index is constructed from SPX option prices, but the project uses a model-implied VIX proxy derived from expected future variance. For a fixed window  $\Delta$ ,

the model-implied VIX at time  $T$  is defined by

$$\widetilde{\text{VIX}}_T^2 = 100^2 \frac{1}{\Delta} \mathbb{E}_T^{\mathbb{Q}} \left[ \int_T^{T+\Delta} v_s ds \right]. \quad (15)$$

The factor  $100^2$  converts variance units into VIX index-point units.

Using the conditional mean (7), we have

$$\frac{1}{\Delta} \mathbb{E}_T^{\mathbb{Q}} \left[ \int_T^{T+\Delta} v_s ds \right] = \frac{1}{\Delta} \int_0^{\Delta} \mathbb{E}_T^{\mathbb{Q}}[v_{T+u}] du \quad (16)$$

$$= \frac{1}{\Delta} \int_0^{\Delta} [\theta + (v_T - \theta)e^{-\kappa u}] du. \quad (17)$$

Evaluating the integral gives

$$\frac{1}{\Delta} \mathbb{E}_T^{\mathbb{Q}} \left[ \int_T^{T+\Delta} v_s ds \right] = \theta + (v_T - \theta) \frac{1 - e^{-\kappa \Delta}}{\kappa \Delta}. \quad (18)$$

Define

$$B_{\Delta} = \frac{1 - e^{-\kappa \Delta}}{\kappa \Delta}, \quad (19)$$

$$A_{\Delta} = \theta(1 - B_{\Delta}). \quad (20)$$

Then the model-implied VIX proxy can be written as

$$\boxed{\widetilde{\text{VIX}}_T = 100 \sqrt{A_{\Delta} + B_{\Delta} v_T}}. \quad (21)$$

The relation in (21) is central for both the Monte Carlo and COS pricing engines. In Monte Carlo, terminal variance is simulated path by path and then converted into VIX. In the COS method, the payoff becomes a deterministic function of terminal variance, which allows the terminal CIR characteristic function to be used.

### 3.7 Terminal CIR Distribution

For the COS pricing engine, the relevant state variable is terminal variance  $v_T$ . The CIR process has a known terminal distribution:

$$v_T \sim c_T \chi_d^2(\lambda_T), \quad (22)$$

where

$$c_T = \frac{\sigma_v^2(1 - e^{-\kappa T})}{4\kappa}, \quad (23)$$

$$d = \frac{4\kappa\theta}{\sigma_v^2}, \quad (24)$$

$$\lambda_T = \frac{4\kappa e^{-\kappa T} v_0}{\sigma_v^2(1 - e^{-\kappa T})}. \quad (25)$$

Here,  $d$  is the degrees-of-freedom parameter and  $\lambda_T$  is the noncentrality parameter.

The characteristic function of  $v_T$  is therefore

$$\boxed{\varphi_{v_T}(u) = (1 - 2iuc_T)^{-d/2} \exp\left(\frac{iuc_T\lambda_T}{1 - 2iuc_T}\right)}. \quad (26)$$

This is different from the transform of integrated variance. The distinction is important because the variance swap depends on  $\int_0^T v_s ds$ , while the model-implied VIX payoff in this project depends on terminal variance  $v_T$  through (21).

### 3.8 Pricing Implications

The formulas above determine the roles of the two numerical engines. The Monte Carlo engine is used when path simulation is natural, particularly for realised variance. The COS engine is used when the payoff can be written as a function of terminal variance and the characteristic function is available.

For a model-implied VIX call with strike  $K$ , the payoff is

$$g_{\text{call}}(v_T) = \left(100\sqrt{A_\Delta + B_\Delta v_T} - K\right)^+. \quad (27)$$

This payoff can be priced either by Monte Carlo simulation or by the COS method. The agreement between the two pricing approaches provides a consistency check for the implementation.

## 4 Monte Carlo Framework

This section describes the Monte Carlo component of the project. The purpose of the W3 implementation is to simulate the Heston stochastic volatility model, estimate realised variance from simulated asset paths, validate the simulation against the analytical Heston variance swap strike, and price a model-implied VIX call option. Monte Carlo simulation is used because it provides a flexible numerical approximation of risk-neutral expectations and can be applied directly to path-dependent quantities such as realised variance (Hirsa, 2024).

The Monte Carlo engine also serves as an independent benchmark for the Fourier-cosine pricing method developed in Section 5. While the COS method is deterministic and based on the terminal variance characteristic function, Monte Carlo uses simulated paths and therefore contains sampling and discretisation error. Comparing the two methods under the same parameter set provides a useful internal validation check.

## 4.1 Simulation Objective

The Monte Carlo implementation has three main objectives. First, it generates joint paths of the asset price  $S_t$  and variance  $v_t$  under the risk-neutral Heston dynamics. Second, it computes realised variance from simulated log returns and compares the average realised variance with the analytical variance swap strike. Third, it prices a model-implied VIX call option by transforming terminal variance into a VIX proxy and averaging discounted payoffs.

The simulation is performed under the risk-neutral measure  $\mathbb{Q}$ , because derivative prices are represented as discounted risk-neutral expectations. For a terminal payoff  $\Phi$ , the general pricing relation is

$$V_0 = e^{-rT} \mathbb{E}^{\mathbb{Q}}[\Phi]. \quad (28)$$

Monte Carlo approximates this expectation by simulating  $N$  independent paths and computing the sample average:

$$V_0^{MC} = e^{-rT} \frac{1}{N} \sum_{j=1}^N \Phi^{(j)}. \quad (29)$$

## 4.2 Heston Path Simulation

The simulated model is the risk-neutral Heston system:

$$\frac{dS_t}{S_t} = r dt + \sqrt{v_t} dW_t^S, \quad (30)$$

$$dv_t = \kappa(\theta - v_t) dt + \sigma_v \sqrt{v_t} dW_t^v, \quad (31)$$

$$d\langle W^S, W^v \rangle_t = \rho dt. \quad (32)$$

The correlation between the Brownian motions is implemented by generating two independent standard normal variables  $Z_v$  and  $Z_{\perp}$ , and then defining

$$Z_S = \rho Z_v + \sqrt{1 - \rho^2} Z_{\perp}. \quad (33)$$

This construction ensures that

$$\text{Corr}(Z_S, Z_v) = \rho. \quad (34)$$

The time interval  $[0, T]$  is divided into  $m$  equal steps of length

$$\Delta t = \frac{T}{m}. \quad (35)$$

The simulation stores both the stock paths and the variance paths. The stock paths are used to compute realised variance, while the terminal variance values are used for model-implied VIX pricing.

### 4.3 Full-Truncation Euler Scheme

The variance process contains the square-root term  $\sqrt{v_t}$ . In continuous time, the CIR process is non-negative under standard conditions. However, in discrete-time numerical simulation, a simple Euler update may produce negative variance values. Since  $\sqrt{v_t}$  is not defined for negative  $v_t$ , a stability correction is required.

The implementation uses a full-truncation Euler scheme. Define

$$v_t^+ = \max(v_t, 0). \quad (36)$$

The variance update is

$$v_{t+\Delta t} = v_t + \kappa(\theta - v_t^+)\Delta t + \sigma_v \sqrt{v_t^+} \sqrt{\Delta t} Z_v. \quad (37)$$

After the update, the next variance value is floored at zero:

$$v_{t+\Delta t} \leftarrow \max(v_{t+\Delta t}, 0). \quad (38)$$

The stock price is simulated in logarithmic form:

$$\log S_{t+\Delta t} = \log S_t + \left(r - \frac{1}{2}v_t^+\right) \Delta t + \sqrt{v_t^+} \sqrt{\Delta t} Z_S. \quad (39)$$

The log update preserves positivity of the simulated stock price and is consistent with the diffusion form of the asset dynamics.

The simulation of square-root processes requires careful treatment near the zero boundary. Recent work on Heston and related square-root processes emphasises that discretisation choices can materially affect pricing accuracy, especially for variance-sensitive products (Choi and Kwok, 2024; Abi Jaber, 2024). The full-truncation scheme is therefore used as a practical balance between numerical stability and implementation simplicity.

## 4.4 Realised Variance Estimation

After simulating stock paths, realised variance is computed from discrete log returns. For a path observed at time points

$$0 = t_0 < t_1 < \dots < t_m = T,$$

the realised variance estimator is

$$\widehat{\text{RV}}_{0,T}^{(j)} = \frac{1}{T} \sum_{i=1}^m \left( \log \frac{S_{t_i}^{(j)}}{S_{t_{i-1}}^{(j)}} \right)^2, \quad (40)$$

where  $j$  indexes the simulated path.

The Monte Carlo estimate of expected realised variance is the sample mean:

$$\overline{\text{RV}}_{0,T}^{MC} = \frac{1}{N} \sum_{j=1}^N \widehat{\text{RV}}_{0,T}^{(j)}. \quad (41)$$

As the time discretisation becomes finer, realised variance approximates average integrated variance:

$$\widehat{\text{RV}}_{0,T}^{(j)} \approx \frac{1}{T} \int_0^T v_s^{(j)} ds. \quad (42)$$

Therefore, the Monte Carlo mean should approximate the analytical variance swap strike:

$$\overline{\text{RV}}_{0,T}^{MC} \approx K_{\text{var}}(0, T). \quad (43)$$

This comparison is the first validation test for the simulation engine. If the Monte Carlo mean realised variance is close to the analytical strike, the simulated variance dynamics are consistent with the theoretical expectation of integrated variance.

## 4.5 Variance Swap Benchmark

The analytical Heston variance swap strike is

$$K_{\text{var}}(0, T) = \theta + (v_0 - \theta) \frac{1 - e^{-\kappa T}}{\kappa T}. \quad (44)$$

The Monte Carlo engine compares this value with the mean realised variance. The error can be reported as an absolute error,

$$\text{AbsError}_{RV} = \left| \overline{\text{RV}}_{0,T}^{MC} - K_{\text{var}}(0, T) \right|, \quad (45)$$



or as a relative error,

$$\text{RelError}_{RV} = \frac{|\overline{\text{RV}}_{0,T}^{MC} - K_{\text{var}}(0, T)|}{K_{\text{var}}(0, T)}. \quad (46)$$

This benchmark is useful because it checks the numerical implementation against a closed-form theoretical quantity. It does not require market data and can therefore be performed before calibration.

## 4.6 Model-Implied VIX Call Pricing

The second pricing task in the Monte Carlo engine is a model-implied VIX call. The project uses the Heston-implied VIX proxy

$$\widetilde{\text{VIX}}_T = 100\sqrt{A_\Delta + B_\Delta v_T}. \quad (47)$$

For each simulated path  $j$ , terminal variance  $v_T^{(j)}$  is transformed into terminal model-implied VIX:

$$\widetilde{\text{VIX}}_T^{(j)} = 100\sqrt{A_\Delta + B_\Delta v_T^{(j)}}. \quad (48)$$

The call payoff is

$$\Phi^{(j)} = \left( \widetilde{\text{VIX}}_T^{(j)} - K \right)^+. \quad (49)$$

The Monte Carlo call price is then

$$C_0^{MC}(K, T) = e^{-rT} \frac{1}{N} \sum_{j=1}^N \left( 100\sqrt{A_\Delta + B_\Delta v_T^{(j)}} - K \right)^+. \quad (50)$$

This pricing method is straightforward because it only requires terminal variance values. However, since the terminal variance values come from discretised Heston paths, the result may contain both sampling error and time-discretisation bias.

## 4.7 Monte Carlo Confidence Interval

The uncertainty of the Monte Carlo estimate is measured by the standard error. Let

$$Y_j = e^{-rT} \left( 100\sqrt{A_\Delta + B_\Delta v_T^{(j)}} - K \right)^+$$

be the discounted payoff on path  $j$ . The Monte Carlo price is

$$C_0^{MC} = \frac{1}{N} \sum_{j=1}^N Y_j. \quad (51)$$

The sample standard deviation is

$$s_Y = \sqrt{\frac{1}{N-1} \sum_{j=1}^N (Y_j - C_0^{MC})^2}. \quad (52)$$

The standard error is

$$SE(C_0^{MC}) = \frac{s_Y}{\sqrt{N}}. \quad (53)$$

An approximate 95% confidence interval is

$$C_0^{MC} \pm 1.96 SE(C_0^{MC}). \quad (54)$$

This interval helps interpret the difference between Monte Carlo and COS prices. A small difference between the two methods is meaningful only when the Monte Carlo uncertainty is sufficiently controlled.

## 4.8 Convergence Behaviour

Monte Carlo convergence is governed by the central limit theorem. The sampling error decreases at the rate

$$O(N^{-1/2}), \quad (55)$$

where  $N$  is the number of simulated paths. This rate is independent of the dimension of the simulated process, which is one reason Monte Carlo is widely used in finance. However, the convergence is relatively slow: reducing the standard error by a factor of two requires approximately four times as many paths.

The implementation therefore studies price stability across different path counts. In addition to increasing  $N$ , time-discretisation error can be reduced by increasing the number of time steps  $m$ . In practice, both sources of error must be considered. A large number of paths does not eliminate bias caused by a coarse time grid.

## 4.9 Interpretation of W3 Results

The W3 component provides a flexible simulation-based pricing framework. Its first validation step is the comparison between simulated realised variance and the analytical variance swap strike. Its second output is a Monte Carlo price for a model-implied VIX call. These two tasks show that the engine can be used both for path-based quantities and terminal-payoff derivatives.

The Monte Carlo engine is not intended to be the fastest pricing method in the project. Instead, its role is to provide a transparent and flexible benchmark. It is especially useful because the simulated paths can later be reused for additional tasks such as scenario

analysis, preliminary variance-risk comparisons, and empirical extensions once market data are added.

## 4.10 Limitations of the Monte Carlo Engine

The current Monte Carlo implementation has several limitations. First, it uses an Euler-type discretisation rather than exact simulation of the CIR process. This is practical and stable, but it may introduce discretisation bias. Second, the current implementation uses synthetic parameters rather than calibrated market parameters. Third, the VIX quantity priced in the simulation is a Heston-implied proxy rather than the official Cboe VIX index.

These limitations do not invalidate the W3 implementation. Rather, they define the scope of the current stage. The goal of W3 is to build and validate a simulation engine under controlled model assumptions. Future stages can extend this engine by adding market calibration, exact CIR sampling, variance risk premium backtesting, and real-data validation.

# 5 CIR-Based COS Pricing Framework

This section presents the Fourier-cosine pricing component of the project. The purpose of the W4 implementation is to provide a deterministic semi-analytical pricing engine for model-implied VIX derivatives. Unlike Monte Carlo simulation, which estimates expectations through random sampling, the COS method approximates the pricing integral using the characteristic function of the relevant state variable. In this project, the state variable is the terminal CIR variance  $v_T$ .

Fourier-based pricing methods are useful when the characteristic function is available even if the transition density is not directly used in closed form (Hirsa, 2024). The COS method is particularly attractive because it approximates the density and expected payoff through a finite cosine expansion. Recent work on the COS method emphasises that both the truncation interval and the number of cosine terms are central numerical choices for pricing accuracy (Junike and Pankrashkin, 2022; Junike, 2024).

## 5.1 Purpose of the COS Engine

The COS engine has three main purposes in the project. First, it provides a pricing method for VIX call and put options under the model-implied VIX proxy. Second, it gives a deterministic benchmark for the Monte Carlo VIX call price from Section 4. Third, it supports additional validation through density recovery and convergence checks.

The method is well suited to the present setting because the terminal variance  $v_T$  has a known characteristic function under the CIR process. Since the model-implied VIX is

written as

$$\widetilde{\text{VIX}}_T = 100\sqrt{A_\Delta + B_\Delta v_T}, \quad (56)$$

a VIX derivative payoff can be treated as a deterministic function of terminal variance. This allows pricing to be formulated as an expectation over  $v_T$ .

## 5.2 Terminal CIR Characteristic Function

The variance process in the Heston model follows the CIR dynamics

$$dv_t = \kappa(\theta - v_t)dt + \sigma_v\sqrt{v_t}dW_t^v. \quad (57)$$

For maturity  $T$ , the terminal variance has the scaled non-central chi-square representation

$$v_T \sim c_T \chi_d^2(\lambda_T), \quad (58)$$

where

$$c_T = \frac{\sigma_v^2(1 - e^{-\kappa T})}{4\kappa}, \quad (59)$$

$$d = \frac{4\kappa\theta}{\sigma_v^2}, \quad (60)$$

$$\lambda_T = \frac{4\kappa e^{-\kappa T}v_0}{\sigma_v^2(1 - e^{-\kappa T})}. \quad (61)$$

The corresponding characteristic function is

$$\boxed{\varphi_{v_T}(u) = (1 - 2iuc_T)^{-d/2} \exp\left(\frac{iuc_T\lambda_T}{1 - 2iuc_T}\right)}. \quad (62)$$

This is the characteristic function used in the COS pricing engine. It is important to distinguish this terminal variance characteristic function from the transform of integrated variance. Variance swaps depend on the integrated variance  $\int_0^T v_s ds$ , whereas the VIX option payoff in this implementation depends on terminal variance through the model-implied VIX proxy.

## 5.3 COS Representation of the Density

Let  $X$  be a random variable with density  $f_X(x)$  and characteristic function  $\varphi_X(u)$ . The COS method approximates the density on a finite interval  $[a, b]$  by a Fourier-cosine expansion:

$$f_X(x) \approx \sum_{k=0}^{N-1} A_k \cos\left(\frac{k\pi(x-a)}{b-a}\right), \quad (63)$$

where the prime indicates that the first term is multiplied by one half. The coefficients are obtained from the characteristic function:

$$A_k = \frac{2}{b-a} \operatorname{Re} \left[ \varphi_X \left( \frac{k\pi}{b-a} \right) e^{-ik\pi a/(b-a)} \right]. \quad (64)$$

In this project,  $X = v_T$ , so  $\varphi_X = \varphi_{v_T}$ . The approximation in (63) is therefore applied to the terminal CIR variance density. This density recovery is not only useful for interpretation, but also serves as a numerical check that the characteristic function and truncation interval are implemented consistently.

## 5.4 COS Pricing Formula

For a derivative payoff  $g(X)$ , the time-zero price under the risk-neutral measure is

$$V_0 = e^{-rT} \mathbb{E}^{\mathbb{Q}}[g(X)]. \quad (65)$$

Using the COS density approximation, this expectation can be approximated as

$$V_0 \approx e^{-rT} \sum_{k=0}^{N-1} \operatorname{Re} \left[ \varphi_X \left( \frac{k\pi}{b-a} \right) e^{-ik\pi a/(b-a)} \right] G_k, \quad (66)$$

where

$$G_k = \frac{2}{b-a} \int_a^b g(x) \cos \left( \frac{k\pi(x-a)}{b-a} \right) dx \quad (67)$$

is the payoff cosine coefficient.

The expression in (66) is the main pricing formula used by the COS engine. It separates the distributional part, represented by the characteristic function, from the payoff part, represented by  $G_k$ . This separation is one of the reasons Fourier-based methods are efficient for repeated pricing across different strikes.

## 5.5 VIX Call and Put Payoffs

In the current implementation, the model-implied VIX proxy is

$$\widetilde{\text{VIX}}_T = 100 \sqrt{A_\Delta + B_\Delta v_T}. \quad (68)$$

Therefore, the payoff of a VIX call with strike  $K$  can be written as a function of terminal variance:

$$g_{\text{call}}(v) = \left( 100 \sqrt{A_\Delta + B_\Delta v} - K \right)^+. \quad (69)$$

Similarly, the payoff of a VIX put is

$$g_{\text{put}}(v) = \left(K - 100\sqrt{A_{\Delta} + B_{\Delta}v}\right)^+. \quad (70)$$

The call and put prices are then computed by substituting these payoff functions into the COS pricing formula. For the call,

$$C_0^{COS}(K, T) \approx e^{-rT} \sum_{k=0}^{N-1} \text{Re} \left[ \varphi_{v_T} \left( \frac{k\pi}{b-a} \right) e^{-ik\pi a/(b-a)} \right] G_k^{\text{call}}, \quad (71)$$

where

$$G_k^{\text{call}} = \frac{2}{b-a} \int_a^b \left(100\sqrt{A_{\Delta} + B_{\Delta}v} - K\right)^+ \cos \left( \frac{k\pi(v-a)}{b-a} \right) dv. \quad (72)$$

The put price is computed analogously:

$$P_0^{COS}(K, T) \approx e^{-rT} \sum_{k=0}^{N-1} \text{Re} \left[ \varphi_{v_T} \left( \frac{k\pi}{b-a} \right) e^{-ik\pi a/(b-a)} \right] G_k^{\text{put}}. \quad (73)$$

In the current implementation, the payoff coefficients are computed numerically. This is practical because the VIX payoff contains a square-root transformation of terminal variance. Closed-form payoff coefficients may be considered in future work to improve computational efficiency.

## 5.6 Truncation Interval

A central practical issue in the COS method is the selection of the truncation interval  $[a, b]$ . Since CIR variance is non-negative, the lower bound is chosen as

$$a = 0. \quad (74)$$

The upper bound is selected using the mean and standard deviation of terminal variance:

$$b = \mathbb{E}^{\mathbb{Q}}[v_T] + L\sqrt{\text{Var}^{\mathbb{Q}}(v_T)}. \quad (75)$$

Here,  $L > 0$  controls the width of the interval. In the implementation,  $L = 8$  is used as a practical benchmark setting.

The truncation interval is a critical input in the COS method. If the interval is too narrow, relevant probability mass may be excluded. If the interval is too wide, the approximation may become less efficient because the cosine expansion must represent the function over a larger range. For this reason, recent COS-method literature treats interval selection and the number of expansion terms as key accuracy controls (Junike

and Pankrashkin, 2022; Junike, 2024).

## 5.7 Density Recovery Check

The COS engine also includes a density recovery check. Using the coefficients in (64), the recovered terminal variance density is

$$\hat{f}_{v_T}(v) = \sum_{k=0}^{N-1} A_k \cos\left(\frac{k\pi(v-a)}{b-a}\right). \quad (76)$$

A basic consistency check is that the recovered density integrates approximately to one:

$$\int_a^b \hat{f}_{v_T}(v) dv \approx 1. \quad (77)$$

The first moment can also be compared with the analytical CIR mean:

$$\mathbb{E}^{\mathbb{Q}}[v_T] = \theta + (v_0 - \theta)e^{-\kappa T}. \quad (78)$$

These checks are important because the COS method relies on both the characteristic function and the truncation interval. A pricing result may appear numerically reasonable even if the recovered density is poorly normalised. Density recovery provides an additional diagnostic tool before interpreting option prices.

## 5.8 VIX Futures Term Structure

The COS framework can also be used to compute a model-implied VIX futures level. Since a futures level is an expected future index level under the risk-neutral measure, the model-implied VIX futures value is

$$F(0, T) = \mathbb{E}^{\mathbb{Q}} \left[ 100 \sqrt{A_{\Delta} + B_{\Delta} v_T} \right]. \quad (79)$$

There is no discount factor in this expression because a futures price is quoted as a forward level rather than as a discounted payoff.

The VIX term structure reflects mean reversion in the CIR variance process. If  $v_0 > \theta$ , variance is expected to decline toward its long-run level, producing a downward-sloping curve. This corresponds to backwardation. If  $v_0 < \theta$ , variance is expected to rise toward its long-run level, producing an upward-sloping curve. This corresponds to contango.

## 5.9 Convergence with Respect to COS Terms

The number of COS terms  $N$  controls the resolution of the cosine expansion. A small value of  $N$  may produce a poor approximation, while a very large value increases computation without necessarily improving the result if truncation error dominates. Therefore, the implementation studies convergence by computing prices for increasing values of  $N$ .

The expected behaviour is that prices stabilise as  $N$  increases. In practice, this stabilisation must be assessed together with the truncation interval. Increasing  $N$  cannot correct an interval that excludes relevant probability mass. For this reason, the convergence study considers both  $N$  and the truncation-width parameter  $L$ .

## 5.10 Comparison with Monte Carlo

The COS price is compared with the Monte Carlo price from Section 4. The absolute difference is

$$\text{AbsError} = |C_0^{COS} - C_0^{MC}|, \quad (80)$$

and the relative difference is

$$\text{RelError} = \frac{|C_0^{COS} - C_0^{MC}|}{C_0^{MC}}. \quad (81)$$

This comparison is useful because the two methods have different sources of numerical error. Monte Carlo contains sampling and time-discretisation error. COS contains truncation error and series approximation error. If both methods produce close prices under the same parameter set, the result supports the internal consistency of the numerical implementation.

## 5.11 Interpretation of W4 Implementation

The W4 implementation extends the project from simulation-based pricing to characteristic-function-based pricing. This is an important methodological step because it introduces a deterministic benchmark that can be used for repeated pricing and validation. The COS engine also makes it possible to recover the terminal variance density and to study the model-implied VIX term structure.

The main practical advantage of the COS method is speed once the characteristic function is available. The main practical challenge is numerical accuracy, especially the selection of the truncation interval and the number of expansion terms. The implementation addresses this by including convergence checks and density recovery diagnostics.



## 5.12 Limitations of the COS Engine

The current COS engine has several limitations. First, the payoff coefficients are computed numerically rather than in closed form. This is acceptable for the current implementation but may be less efficient for large-scale surface construction. Second, the method prices derivatives on a Heston-implied VIX proxy, not on the official Cboe VIX index. Third, the one-factor CIR variance process may not be flexible enough to reproduce real VIX futures and options data after calibration.

These limitations define the natural direction for future work. Possible extensions include closed-form or more efficient payoff coefficient computation, adaptive truncation rules, exact CIR sampling for validation, and multi-factor or rough-volatility specifications for better market fit.

## 6 Results and Discussion

This section discusses the numerical results obtained from the Monte Carlo and COS pricing engines. The results are based on synthetic model parameters rather than market-calibrated parameters. Therefore, they should be interpreted as internal consistency checks of the numerical framework, not as empirical claims about observed market prices.

The main purpose of the results section is to verify whether the two independently implemented pricing approaches produce consistent outputs under the same model assumptions. The Monte Carlo engine simulates Heston paths and estimates payoffs by averaging discounted simulated values. The COS engine uses the characteristic function of terminal CIR variance and evaluates the same model-implied VIX payoff through a Fourier-cosine expansion. Agreement between these methods supports the correctness of the numerical implementation.

### 6.1 Benchmark Parameter Set

The comparison between Monte Carlo and COS pricing is performed under the following parameter set:

$$v_0 = 0.04, \quad \kappa = 2, \quad \theta = 0.04, \quad \sigma_v = 0.5, \quad (82)$$

with option maturity

$$T = \frac{30}{365}, \quad (83)$$

risk-free rate

$$r = 0.03, \quad (84)$$

and VIX option strike

$$K = 20. \quad (85)$$

The values  $v_0 = 0.04$  and  $\theta = 0.04$  imply that the initial variance is equal to the long-run variance. In volatility-point terms, this corresponds approximately to a long-run volatility level of 20%, because  $\sqrt{0.04} = 0.20$ . This parameter setting is useful for benchmark testing because the model starts near its long-run variance level and does not impose a strong upward or downward mean-reversion effect at the initial time.

## 6.2 Monte Carlo Validation through Realised Variance

The first validation step in W3 is the comparison between simulated realised variance and the analytical Heston variance swap strike. The realised variance is computed path by path as

$$\widehat{\text{RV}}_{0,T}^{(j)} = \frac{1}{T} \sum_{i=1}^m \left( \log \frac{S_{t_i}^{(j)}}{S_{t_{i-1}}^{(j)}} \right)^2. \quad (86)$$

The analytical benchmark is

$$K_{\text{var}}(0, T) = \theta + (v_0 - \theta) \frac{1 - e^{-\kappa T}}{\kappa T}. \quad (87)$$

When the average Monte Carlo realised variance is close to  $K_{\text{var}}(0, T)$ , the simulation is consistent with the theoretical expectation of integrated variance. This is an important test because realised variance is computed from simulated stock paths, whereas the analytical strike is derived from the variance process. Agreement between the two confirms that the joint simulation of stock and variance paths behaves consistently with the model.

This test is also useful because it does not require market data. Before calibrating the model to real prices, the implementation can be checked against a known theoretical quantity. In this sense, the variance swap comparison functions as a unit-level validation of the Monte Carlo engine.

## 6.3 Monte Carlo VIX Call Price

The Monte Carlo VIX call price is obtained by transforming terminal variance into the model-implied VIX proxy:

$$\widetilde{\text{VIX}}_T = 100 \sqrt{A_\Delta + B_\Delta v_T}. \quad (88)$$

The discounted payoff is then averaged across simulated paths:

$$C_0^{MC} = e^{-rT} \frac{1}{N} \sum_{j=1}^N \left( 100 \sqrt{A_\Delta + B_\Delta v_T^{(j)}} - K \right)^+. \quad (89)$$

Under the benchmark parameter set, the Monte Carlo estimate is

$$C_0^{MC} \approx 1.984952. \quad (90)$$

This value is simulation-based and therefore depends on the number of paths, the time discretisation, and the random seed. It is not expected to be exactly equal to the COS price. Instead, the Monte Carlo result should be interpreted together with its sampling uncertainty and convergence behaviour.

## 6.4 COS VIX Call Price

The COS method prices the same model-implied VIX call using the terminal CIR characteristic function. The payoff is still

$$g(v) = \left(100\sqrt{A_\Delta + B_\Delta v} - K\right)^+, \quad (91)$$

but the expected payoff is computed through a Fourier-cosine expansion rather than through simulated paths.

Under the same benchmark parameter set, the COS price is

$$C_0^{COS} \approx 2.008969. \quad (92)$$

Unlike Monte Carlo, this value is deterministic for a fixed number of COS terms, truncation interval, and integration grid used for payoff coefficients. Therefore, the COS result provides a useful semi-analytical benchmark for the simulation-based estimate.

## 6.5 Comparison between Monte Carlo and COS

The numerical comparison is summarised as follows:

| Quantity            | Value    | Interpretation                           |
|---------------------|----------|------------------------------------------|
| COS price           | 2.008969 | Characteristic-function-based estimate   |
| Monte Carlo price   | 1.984952 | Simulation-based estimate                |
| Absolute difference | 0.024017 | Difference between the two engines       |
| Relative difference | 1.21%    | Difference relative to Monte Carlo price |

The absolute difference is

$$\text{AbsError} = |C_0^{COS} - C_0^{MC}| = 0.024017. \quad (93)$$

The relative difference is

$$\text{RelError} = \frac{|C_0^{COS} - C_0^{MC}|}{C_0^{MC}} \approx 1.21\%. \quad (94)$$

This result supports the numerical consistency of the two pricing engines. Exact equality is not expected because the two methods have different sources of numerical error. Monte Carlo contains sampling error and time-discretisation bias. The COS method contains truncation error, finite-series approximation error, and numerical integration error in the payoff coefficients. A relative difference of approximately 1.21% is therefore acceptable for an internal benchmark at this stage of the project.

## 6.6 Interpretation of the Pricing Difference

The observed difference between the COS and Monte Carlo prices should be interpreted carefully. A small difference does not prove that both implementations are mathematically perfect, but it provides evidence that the two methods are broadly aligned under the same assumptions.

The Monte Carlo estimate is affected by random sampling. Increasing the number of paths should reduce the sampling component of the error at the rate  $O(N^{-1/2})$ . However, increasing the number of paths alone does not remove time-discretisation bias. If the time grid is too coarse, the simulated variance and stock paths may still deviate from the continuous-time model.

The COS estimate, by contrast, is not affected by random sampling. Its accuracy depends mainly on the number of cosine terms  $N$ , the truncation interval  $[a, b]$ , and the numerical computation of payoff coefficients. If the truncation interval is too narrow, the method may exclude relevant probability mass. If the interval is too wide, the approximation may become less efficient.

Therefore, the comparison between COS and Monte Carlo should be viewed as a cross-validation exercise. The two methods are based on different numerical principles. When they produce close results, confidence in the implementation increases.

## 6.7 COS Density Recovery

The COS engine also performs density recovery for terminal CIR variance. The recovered density is computed using the same characteristic function that is used for pricing:

$$\hat{f}_{v_T}(v) = \sum_{k=0}^{N-1} A_k \cos\left(\frac{k\pi(v-a)}{b-a}\right). \quad (95)$$

A basic diagnostic check is whether the recovered density integrates approximately to one:

$$\int_a^b \hat{f}_{v_T}(v) dv \approx 1. \quad (96)$$

This check is important because a pricing output can appear plausible even if the underlying density approximation is poor. Density recovery provides a more direct test of the characteristic function, the truncation interval, and the cosine expansion. If the recovered density fails to integrate close to one, then option prices based on the same expansion may not be reliable.

## 6.8 Convergence in the Number of COS Terms

The COS convergence study evaluates how the price changes as the number of expansion terms  $N$  increases. A stable COS implementation should show that prices become less sensitive to  $N$  after a sufficiently large number of terms. In the benchmark implementation,  $N = 128$  is used as the reported setting.

The number of terms should not be interpreted independently from the truncation interval. Increasing  $N$  improves the resolution of the cosine expansion within the chosen interval, but it does not fix an interval that excludes important probability mass. This is why the W4 validation framework studies both the number of terms and the truncation-width parameter.

## 6.9 Convergence in the Truncation Width

The truncation interval is set as

$$a = 0, \quad b = \mathbb{E}^{\mathbb{Q}}[v_T] + L\sqrt{\text{Var}^{\mathbb{Q}}(v_T)}. \quad (97)$$

The lower bound is zero because CIR variance is non-negative. The upper bound depends on the truncation-width parameter  $L$ .

The role of  $L$  is to include most of the relevant probability mass of terminal variance. If  $L$  is too small, tail values of  $v_T$  may be excluded. If  $L$  is too large, the cosine expansion is performed over an unnecessarily wide interval. The implementation uses  $L = 8$  as a practical benchmark and studies sensitivity to alternative values.

## 6.10 Model-Implied VIX Term Structure

The W4 implementation also computes a model-implied VIX futures term structure:

$$F(0, T) = \mathbb{E}^{\mathbb{Q}} \left[ 100\sqrt{A_{\Delta} + B_{\Delta}v_T} \right]. \quad (98)$$

This quantity is a futures level, not a discounted derivative price. Therefore, no discount factor appears in the expression.

The shape of the model-implied term structure follows from mean reversion in the CIR variance process. When  $v_0 > \theta$ , current variance is above its long-run mean, so expected future variance decreases with maturity. This produces a downward-sloping VIX futures curve, associated with backwardation. When  $v_0 < \theta$ , current variance is below its long-run mean, so expected future variance increases with maturity. This produces an upward-sloping curve, associated with contango.

This result demonstrates that even a simple one-factor variance process can reproduce basic mean-reversion patterns in the VIX term structure. However, it does not imply that the one-factor model can fit real market VIX futures data accurately across all maturities. Market calibration is left for later stages.

## 6.11 Testing and Completeness

The implementation includes tests for several mathematical and numerical components. These include parameter validation, characteristic function boundary conditions, VIX affine mapping, terminal density recovery, non-negativity of option prices, VIX futures positivity, and convergence output structure.

The purpose of these tests is not only to verify that functions run without errors. They also check mathematical properties that should hold regardless of the chosen parameter set. For example, the characteristic function should satisfy

$$\varphi(0) = 1, \tag{99}$$

and for real  $u$ ,

$$|\varphi(u)| \leq 1. \tag{100}$$

Similarly, the recovered density should be non-negative after numerical correction and should integrate approximately to one.

These checks help ensure that the implementation is mathematically consistent and not merely producing numerical outputs.

## 6.12 Overall Discussion

The W3 and W4 results show that the project has a working numerical core. The Monte Carlo engine provides a flexible simulation-based method for realised variance and VIX call pricing. The COS engine provides a faster deterministic method based on the terminal variance characteristic function. Their comparison gives a meaningful cross-validation framework.

The current results should be interpreted within the scope of a model-based numerical project. They are not market-calibrated prices, and they do not yet constitute an empirical volatility risk premium study. Instead, they demonstrate that the theoretical formulas, code implementation, and numerical validation are aligned.

This stage prepares the project for the next components: real data collection, model calibration, comparison with market prices, and volatility risk premium backtesting.

## 7 Conclusion

This project studied volatility as a tradable asset class through variance swaps, model-implied VIX derivatives, and numerical pricing under stochastic volatility. The main objective was to connect the theoretical structure of volatility-linked products with working numerical methods that can later be extended to real market data, empirical calibration, hedging analysis, and volatility risk premium backtesting.

The first part of the project reviewed the role of volatility derivatives in modern financial markets. Variance swaps, VIX futures, and VIX options provide direct exposure to volatility and variance rather than only to the direction of the underlying asset. This motivates a modelling framework in which variance is treated as a stochastic state variable. The Heston model was used as the baseline framework because its variance process is mean-reverting, non-negative in continuous time, and analytically tractable enough for both simulation and characteristic-function-based pricing.

The theoretical part derived the key formulas required for the numerical implementation. The fair variance swap strike was obtained from the expected integrated variance under the Heston/CIR variance process. The project also constructed a model-implied VIX proxy based on the conditional expectation of future variance. This distinction is important because the proxy used in the project is not the official Cboe VIX index. Instead, it is a Heston-implied quantity of the form

$$\widetilde{\text{VIX}}_T = 100\sqrt{A_\Delta + B_\Delta v_T}.$$

This representation allowed VIX-type payoffs to be expressed as functions of terminal variance.

The W3 component implemented the Monte Carlo engine. It simulated Heston stock and variance paths using a full-truncation Euler scheme, computed realised variance from squared log returns, and compared the simulated realised variance with the analytical Heston variance swap strike. This comparison served as the first validation step for the simulation engine. The same Monte Carlo framework was also used to price a model-implied VIX call by transforming terminal variance into VIX and averaging discounted payoffs.

The W4 component implemented the CIR-based COS pricing engine. The method used the characteristic function of terminal CIR variance and approximated the expected payoff through a Fourier-cosine expansion. This provided a deterministic semi-analytical benchmark for the Monte Carlo VIX call price. The COS framework was also used for density recovery, VIX call and put pricing, VIX futures term-structure calculation, and convergence checks with respect to the number of terms and truncation interval.

The numerical comparison between the Monte Carlo and COS engines showed close agreement under the same parameter set. The reported COS price was approximately 2.008969, while the Monte Carlo price was approximately 1.984952. The absolute difference was 0.024017, corresponding to a relative difference of about 1.21%. This result supports the internal consistency of the two independently implemented pricing methods. The difference is reasonable because Monte Carlo contains sampling and discretisation error, while the COS method contains truncation and finite-series approximation error.

The main contribution of the completed W1–W4 work is therefore a coherent numerical foundation for volatility-linked derivatives. The project now contains a theoretical framework, a simulation-based pricing engine, a characteristic-function-based pricing engine, validation checks, and a modular code structure. These elements prepare the project for later stages involving market data, calibration, hedging analysis, and empirical testing.

Several limitations remain. First, the current results are based on synthetic parameters rather than market-calibrated parameters. Second, the VIX object used in the project is a model-implied proxy, not the official Cboe VIX index. Third, the one-factor Heston/CIR structure may not be flexible enough to reproduce the full SPX volatility surface and VIX term structure simultaneously. Fourth, the current project has not yet performed a full empirical volatility risk premium backtest. These limitations are expected at the current stage and define the natural direction for future work.

Future extensions should focus on the remaining project stages. In W5 and W6, the project should move from synthetic parameters to real market data and calibration. These stages are expected to connect the model to VIX index levels, VIX futures, VIX options, SPX returns, and risk-free rate data. After calibration, the project can compare model prices with market prices and evaluate whether a one-factor variance model is sufficient. If the one-factor Heston/CIR model is not flexible enough, the framework can also be extended to two-factor CIR models or rough volatility specifications.

W7 and W8 should continue the empirical and practical parts of the project. In particular, these later stages can extend the framework toward hedging analysis, volatility risk premium backtesting, final performance evaluation, and interpretation of trading or risk-management implications. The volatility risk premium component should compare implied variance with subsequently realised variance and evaluate whether a systematic volatility strategy can be justified by the model outputs. The hedging component can then assess how model-based prices and sensitivities behave under realistic market movements.



Overall, the completed work establishes the numerical core of the project. The Monte Carlo engine provides flexibility and path-based validation, while the COS engine provides speed and deterministic cross-checking. Together, they form a consistent framework for modelling and pricing volatility-linked derivatives before moving to real data, calibration, hedging, and empirical volatility risk premium analysis.

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